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BRIEF REPORT

Priming the distractor can eliminate the Stroop interference effect

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Abstract

The Stroop interference effect—the slower response to color in an incongruent Stroop stimulus (e.g., **GREEN**) relative to a neutral Stroop stimulus (e.g., **XXXX**) is usually highly robust. The present study investigated the role of selective attention in the Stroop task by priming the distractor word. Replicating previous studies using the verbal (color-naming) task, priming the distractor word produced a substantial speedup of response to the color in a Stroop stimulus in our manual Stroop task. Importantly, priming the distractor completely eliminated the Stroop interference effect (Incongruent = Neutral, e.g., **GREEN** = **XXXX**), and brought about a sizable facilitation effect (Congruent < Neutral, e.g., **RED** < **XXXX**) that was absent in the standard (control-primed) Stroop trials. RT distribution analysis showed that the pattern of facilitation and interference effects was changed radically by priming the distractor: In the standard Stroop task, the Stroop interference effect increased across quantiles, and the facilitation effect was absent throughout the quantiles; in contrast, in the distractor-primed Stroop task, the interference effect was eliminated, and the large facilitation effect that emerged remained constant across the quantiles. We interpret these results in terms of a "Trojan horse" account that suggests that in a Stroop stimulus, color and word form are integrated into an object; hence, when object-based attention is deployed to attend to the color, the word form "sneaks in." Priming the distractor breaks this integration, allowing attention to disengage from the irrelevant word dimension and eliminating Stroop interference.

Keywords Stroop · Selective attention · Stroop interference · Delta plot analyses

In the Stroop color-word task (Stroop, 1935), participants find it challenging to name the color when the stimulus is a word incongruent with the response (e.g., the word GREEN is presented in red font; hereafter, we will denote the color-carrying distractor word in uppercase letters and the font color in lowercase letters, e.g., GREENred). The difficulty can be measured by the considerable slowdown in color naming relative to when the distractor is nonlinguistic symbols (e.g., XXXXred), termed the Stroop interference effect. It is such a robust phenomenon that it has been reported that "nobody fails to Stroop" (Haaf & Rouder, 2019). While it is widely agreed that the Stroop interference effect reflects the failure of selective attention, surprisingly, why it is so difficult "not to Stroop" is unknown. The present study addresses this question.

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At the outset, we emphasize that our interest lies in the "Stroop interference effect," the slower response to the incongruent condition relative to the neutral condition. While the term "Stroop interference" is sometimes used to refer to the difference between the incongruent (e.g., GREENred) and the congruent condition (e.g., GREENgreen; e.g., Haaf & Rouder, 2019), MacLeod (1991) noted that this is not what Stroop (1935) reported, and comparing the incongruent and congruent conditions conflates the Stroop interference and the facilitation effect (the faster response to the congruent condition relative to the neutral condition). The Stroop interference and facilitation effects are not symmetrical (cf. Smith & Ulrich, 2024 "the midpoint assumption"): The facilitation effect is numerically much smaller and less robust than the Stroop interference effect, and the two effects may reflect separate mechanisms (e.g., MacLeod, 1991, 1998). Given this, our discussion will distinguish between Stroop interference and facilitation effects.

The starting point of our research is the view that the Stroop stimulus's integrated nature is a powerful contributor to Stroop interference (MacLeod, 1991), and this means that attending to the color also results in the word receiving

attention (Roelofs & Lamers, 2007). We will first describe research that showed that "breaking of the Gestalt" (integration of color and word form into a single object)—by removing the color from the integrated Stroop stimulus soon after the stimulus onset—facilitates disengagement of attention from the word and results in a reduced Stroop interference effect. We then propose that a manipulation known to facilitate the color naming of a colored word—namely, distractor priming—also facilitates the disengagement of attention from the word in the integrated Stroop object. We will then report an experiment using this manipulation and show that it is possible to eliminate the Stroop interference effect throughout the whole reaction time (RT) distribution.

Roelofs and Lamers (2007) argued that the integrated nature of the Stroop stimulus makes it difficult to deploy spatial or object-based attention to selectively attend to the color (and ignore the word). Attending to the spatial location of the color necessarily results in attending to the word because the color and the word form are spatially contiguous. In object-based attention, all information within the spatial boundaries of the selected object is preferentially processed (e.g., Kahneman & Henik, 1981). In an integrated Stroop stimulus, the color and the word are perceived as constituting two attributes (color and word form) of the same object. Thus, deploying object-based attention to respond to the color does not help to separate the word from the color in a Stroop stimulus either. Instead of aiding, these processes result in a "Trojan Horse" effect: By attending to the colored object, the irrelevant word information "slips in" and is processed unintentionally (Roelofs & Lamers, 2007).

Previous studies (La Heij et al., 2001; Lamers & Roelofs, 2007) have suggested that separating the color and the word in *time* facilitates color naming by "breaking the Gestalt." La Heij et al. (2001) had participants name the color of integrated Stroop stimuli, with the target color removed after 120ms. They found that the Stroop interference effect was reduced relative to a standard Stroop condition in which the color and target remained integrated. Lamers and Roelofs (2007) replicated this finding. They suggested that removing the target color from the integrated Stroop stimulus disrupts the Gestalt grouping by violating the "common fate" of the color and word form attributes, causing participants to no longer see the target and distractor as a single object. They concluded that "if the attributes are not seen as belonging to a single entity, the distractor input is not processed along with the target" (p.1314) and that "temporal segregation of color and word helps to disengage attention from the irrelevant dimension" (Roelofs & Lamers, 2007, p. 136).

We propose that another way to break the integration of color and word in a Stroop stimulus is to present the distractor word in monochrome *before* the integrated Stroop stimulus appears (e.g., GREENblack–GREENred). This manipulation is termed *distractor priming*. Intuitively, priming the distractor word could be expected to slow down target color naming by making the competing irrelevant word more salient. Contrary to this expectation, previous distractor-priming studies have consistently shown that priming the color-carrying distractor word speeds up color naming (Burt, 1994; Chao, 2011; Dyer, 1971; Lamers et al., 2010; McClain, 1983).

However, none of the previous distractor-priming studies discussed breaking the integration of color and word as an explanation for speeded color naming, nor were the studies designed to test whether the Stroop interference/facilitation effect is reduced by distractor priming. Burt (1994) used only color-neutral words (e.g., FAMILY, CHOIR) as the color carriers. Chao (2011) used the incongruent Stroop stimuli in a series of experiments, but only in two experiments did they include the neutral stimuli (Experiment 2) or congruent stimuli (Experiment 3). More critically, Chao did not use a standard Stroop task as their "control precue" condition: Their control precue (referred to as the "invalid prime") was words that did not match the target color or the distractor word (e.g., BLUEblack–GREENred), and the distractor-primed trials (the "valid prime" trials) were presented three times more often than the invalid prime. Dyer (1971) included congruent, incongruent, and neutral Stroop stimuli but employed a 0-ms prime–target SOA block as the standard Stroop condition. In this setup, the integrated Stroop stimulus was not preceded by a prime, so the slower responses in this condition compared to the distractor-primed condition might have resulted from a lack of general preparedness.

Notably, these studies reported only the analysis of mean RTs. Because RT distributions are positively skewed, mean RTs may be an unrepresentative measure of central tendency, and an analysis of mean RTs may overlook theoretically significant findings that are evident only when examining the entire RT distribution (e.g., Balota & Yap, 2011; Heathcote et al., 1991). Next, we briefly review the RT distribution analysis of Stroop effects.

RT distribution analysis

A simple nonparametric approach to analyzing RT distributions is to plot the effects across the quantiles (i.e., delta plots; refer to Methods for more details). A positively sloped delta plot means the effect increases with slower responses. A negatively sloped delta plot means the effect decreases as the responses slow, while a flat delta plot indicates a constant effect across the quantiles (i.e., a distributional shift).

Pratte et al. (2010) noted that in all the previous Stroop experiments they examined, the slope of the delta plot of the Stroop congruency effect (i.e., incongruent minus congruent) was always positive. Pratte et al. pointed out that the

positive delta slope pattern is concordant with information/evidence accumulation models (e.g., the diffusion model of Ratcliff, 1978). In these models, information about a decision (evidence) is accumulated until a criterion is reached, and the positive slope of the delta plot is typically explained in terms of a difference in the rates of evidence accumulation (called the drift rate in the diffusion model) between the conditions. Pratte et al. proposed that the positive-going delta slope in the Stroop task could be explained by the assumption that the information from the distractor word is incorporated into the effective rate of evidence accumulation for the color target.

Kinoshita et al. (2017) included neutral trials to isolate interference and facilitation effects across the RT distribution in a standard and primed version of the Stroop task where the word was presented ahead of colored Xs to be color-named (e.g., GREENblack–XXXXred).[1] In the standard integrated Stroop task, the pattern showed interference dominance, with the large interference effect producing a positive delta slope and the small facilitation effect showing a relatively flat delta slope. The pattern changed completely in the priming Stroop task to facilitation dominance. The now small interference effect produced a flat delta slope, and the large facilitation effect that emerged was also constant across the quantiles. Following Pratte et al. (2010), Kinoshita et al. (2017) interpreted the positive delta slope of the interference effect in the integrated Stroop task as arising from the color and (the incongruent) word information entering the evidence accumulation process simultaneously. The complete absence of an interference effect in the primed version of the Stroop task, in turn, suggests that the integration of word and color into a single object was responsible for the positive delta slope observed. Put differently, when the word and color are temporally segregated, the interference effect (with the positively sloped delta plot) is removed because the incongruent word and color information do not simultaneously enter the evidence accumulation process. In sum, the delta plot patterns can be used to infer how the distractor word impacts the evidence accumulation process.

The present study aimed to test whether the Stroop interference effect can be reduced/eliminated by breaking the integration of the color and word in the Stroop stimulus using distractor priming. Although previous distractor-priming studies have shown consistently that color naming is facilitated, none of these studies assessed the effect of distractor priming on the interference and facilitation effects separately and across the whole RT distribution. To address this gap, the present study designed a distractor-priming paradigm that included congruent, incongruent, and neutral trials, using two types of precue, all mixed randomly, within an experiment. In the distractor-priming ("identity precue") condition, the distractor word was presented before the Stroop stimulus (e.g., GREENwhite–GREENred). In the "control precue" condition that served as the standard Stroop task, an uninformative precue was presented before the Stroop stimulus (e.g., #####white–GREENred). Using a precue–target SOA of 550 ms, we demonstrate that priming the distractor word can eliminate the Stroop interference effect.

[1] Note that this is not the same as the distractor-primed Stroop paradigm, in which the target to be color-named is an integrated Stroop stimulus.

Table 1 Experimental design and example stimuli

Distractor congruency	Precue type	
	Control	Identity
Congruent	##### - RED	RED - RED
Neutral	##### - XXXX	XXXX - XXXX
Incongruent	##### - GREEN	GREEN - GREEN

Note. The examples presented are for the response color red. There were four response colors: red, yellow, green, and blue. See the text for details. (Color table online)

Methods

Participants

Forty students from Macquarie University (five men; mean age = 20 years, $SD = 3.55$) participated in the experiment in return for course credit. The experiment was approved by the Macquarie University Human Research Ethics Committee (reference #520231272546753).

Design and materials

The experiment used a manual Stroop task with four response colors. In this task, stimuli are words, or a series of Xs, presented in one of four colors (red, yellow, green, and blue), and participants were required to identify the color of the stimuli using one of four keys (Z, X, N, or M) on the computer keyboard. Participants were told to press the Z key for red, X for yellow, N for green, and M for blue as fast as possible without making too many errors. The dependent variables were RT (in ms) and error rate.

The experimental design was a 2 (precue type: control or identity) × 3 (distractor congruency: congruent, neutral, incongruent) design, with both factors manipulated within subjects. Table 1 shows examples of stimuli in each condition.

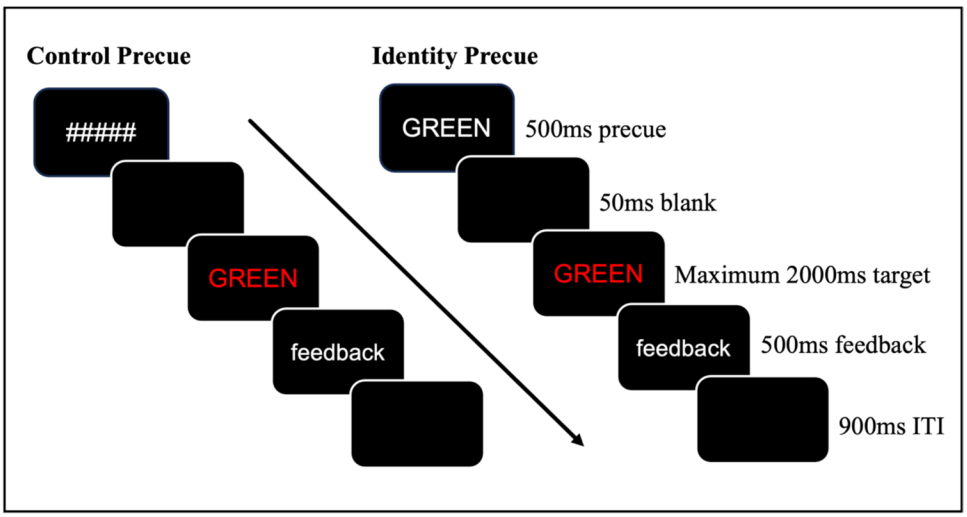


Fig. 1 Example sequence of experimental events. *Note.* Example sequence of events in an incongruent trial for control precue (left) and identity precue (right) conditions. Feedback was given as to whether the word was correct or wrong. ITI = intertrial interval. (Color figure online)

Each trial consisted of a precue followed by the Stroop target to which a response was required. In the control precue condition, the precue was a string of five # signs (#####). This is your standard Stroop. In the identity precue condition, the precue was the same as the distractor (color-carrying) word, presented in white against a black background. There were three distractor congruency types for the Stroop target: congruent, incongruent, and neutral. In the congruent condition, a color name was presented in the matching color (e.g., the word RED in red). In the incongruent condition, a color name was presented in a nonmatching color (e.g., the word RED in green). To keep the color-word contingency equal between the congruent and incongruent conditions, the color name was presented in only one of three nonmatching colors: specifically, the word GREEN was presented in red, YELLOW in blue, RED in green, and BLUE in yellow. In the neutral condition, a string of four Xs (XXXX) was presented in one of four colors. Hence, altogether there were 12 unique Stroop targets (distractor-color combinations), with each of the four colors occurring equally often with the congruent, incongruent, or neutral distractor, and each Stroop target was preceded by the control precue and by the distractor precue (i.e., a set of 24 unique prime–target pairs).

Apparatus and procedure

Stimulus presentation and data collection were achieved with the DMDX display system developed by Forster and Forster (2003). All stimuli and instructions were presented in Arial 14-point font. The stimulus display was synchronized to the screen refresh rate (10.1 ms). A black background was used throughout the experiment, and the instructions and primes were presented in white. RGB (Red, Green, and Blue) values were 255/000/000 for red, 000/000/255 for blue, 000/128/000 for green, and 255/255/000 for yellow. Participants were tested in groups but were seated in individual rooms. Participants were seated approximately 60 cm in front of a computer monitor, upon which stimuli were presented. Each participant first completed a practice block of 20 trials, all with neutral Stroop targets preceded by the control precue (to learn the response color-key assignment). This was followed by a second practice block and eight test blocks of 48 trials (a total of 384 test trials). The second practice block and each test block contained two occurrences of each of the 24 unique precue–target combinations. For each block, a different pseudorandom order of trials was generated with the constraint that the same color did not occur in immediate succession.

At the outset of the experiment, participants were instructed that the task was speeded color identification. They were told that on each trial, they would be presented with a word or symbol in white, followed by a word or symbol in colored font, and their task was to identify the font color by pressing the assigned key as fast as possible without sacrificing accuracy (see Fig. 1 for an illustration of a trial sequence). Each trial started with the presentation of a precue for 500 ms. Following a 50-ms blank interval, a Stroop target was presented for a maximum of 2,000 ms, or until the participant responded, whichever occurred first. Following each response, feedback was given (the word CORRECT or WRONG in white font) during the intertrial interval. The intertrial interval (between the participant's response and the beginning of the subsequent trial) was 1,400 ms.

Statistical analysis

First, we report the mean RTs on correct trials and the error rates[2] in the 2 (precue type: control vs. identity) × 3 (distractor congruency: congruent, neutral, incongruent) design. We

[2] The analysis of error rates can be found in Appendix A.

then report the mean RT as a 2 × 3 fully repeated-measures analysis of variance (ANOVA), followed by planned contrasts to examine the Stroop interference and facilitation effects. These analyses were performed in JAMOVI (Version 2.3.28). Planned contrasts were Bonferroni corrected.

Delta plot analysis

To calculate the quantile estimates, first, correct RTs for each experimental condition were sorted from the fastest to the slowest for each participant, and nine quantile estimates (0.1, 0.2, 0.3, … 0.9) were obtained. The averages of the RTs of the slowest trial of the upper bin and the fastest trial of the lower bin make up the nine observed quantile estimates. Next, quantile plots were averaged over participants per distractor congruency type for each precue type and are presented in Fig. 2. Figure 3 presents delta plots of the interference and facilitation effects in the control and identity precue conditions, corresponding to the difference between two distractor conditions at each quantile, averaged over the participants per condition. In the delta plot, a positively sloped line indicates an increase in the effect across quantiles, a flat line indicates a constant effect across the quantiles, and a negatively sloped line indicates a decrease in the effect across the quantiles.

We then report the analysis of delta plots of interference and facilitation effects using the linear trend of polynomial-trend contrasts on the quantiles factor. Planned polynomial contrasts were performed in SPSS (Version 27.0).

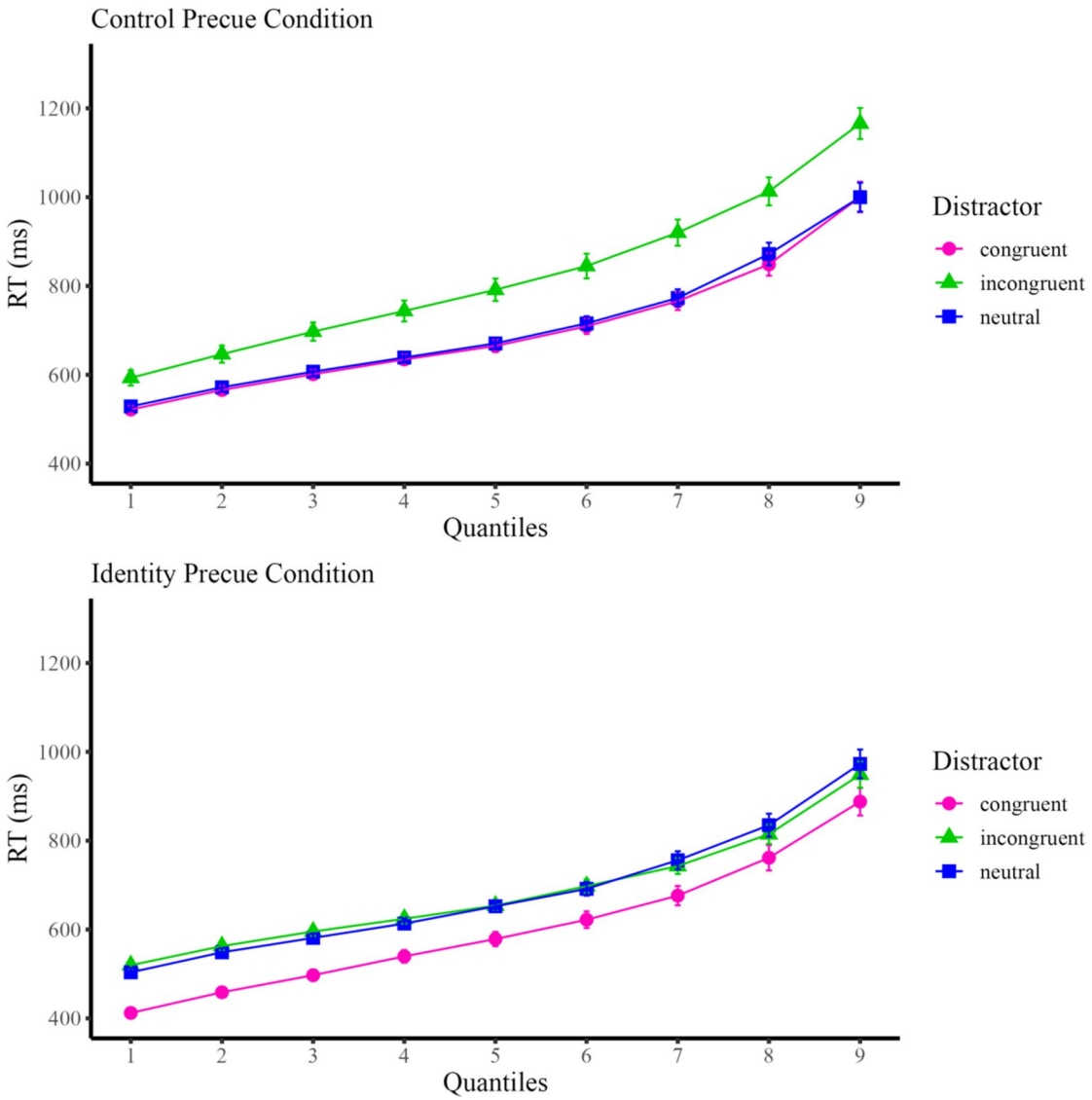


Fig. 2 Quantile plots averaged over participants per distractor congruency type for each precue type. *Note*. Refer to Methods for an explanation of how quantiles are calculated. Error bars are standard errors. (Color figure online)

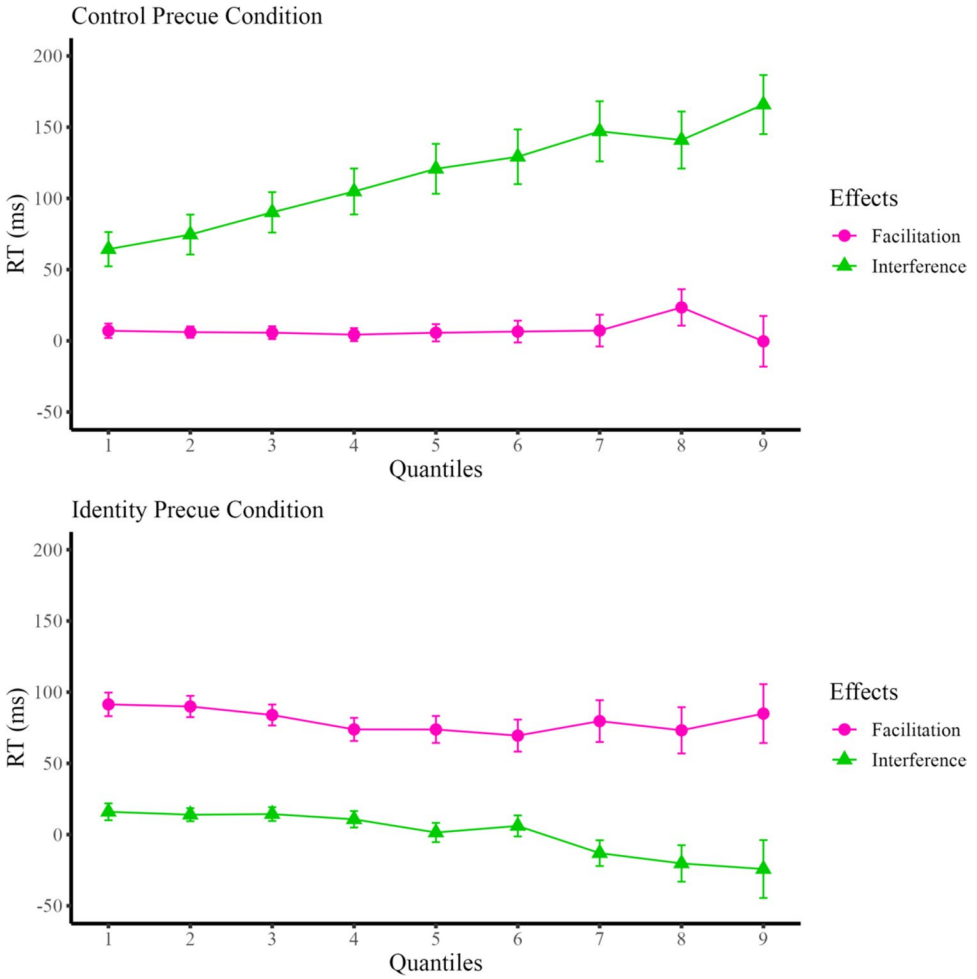


Fig. 3 Delta plots of interference and facilitation effect across quantiles within the standard (control-precue) and distractor-primed Stroop tasks. *Note*. The interference effect (green triangles) is the incongruent condition minus the neutral condition at each quantile, averaged over the participants per precue condition. The facilitation effect (pink circles) is the neutral condition minus the congruent condition at each quantile, averaged over the participants per precue condition. Error bars are standard errors. (Color figure online)

Results

In the analysis of RTs, error trials (642 out of 15,360 data points = 4.18% of data) and RTs faster than 200 ms (five trials = 0.03% of data) were excluded. The mean correct RT and error rates are shown in Table 2.

RT

The 2 (precue) × 3 (distractor congruency) ANOVA showed a significant main effect of precue type, $F(1,39) = 158$, $MSe = 441747$, $p < .001$, $\eta_p^2 = .802$, distractor congruency type, $F(2,78) = 95.6$, $MSe = 203070$, $p < .001$, $\eta_p^2 = .710$, and an interaction, $F(2,78) = 33.5$, $MSe = 66338$, $p < .001$, $\eta_p^2 = .462$. Planned contrasts revealed that participants had faster RTs in the identity precue compared to the control precue condition for all distractor congruency conditions: congruent,

Table 2 Mean and standard deviation (in parentheses) response times (RT, in ms) and percentage error rate (%E)

Distractor congruency	Control precue condition		Identity precue condition	
	RT (*SD*)	%E	RT (*SD*)	%E
Congruent	721 (107)	3.24 (2.88)	625 (109)	3.95 (3.32)
Incongruent	842 (151)	6.41 (5.25)	705 (97)	3.91 (3.28)
Neutral	728 (105)	4.02 (3.50)	704 (101)	3.55 (3.04)
Interference Effect	114 (90)*	2.38 (4.56)	1 (46)	0.35 (3.51)
Facilitation Effect	7 (41)	0.78 (2.72)	79 (60)*	−0.39 (2.98)

Note. Interference effect = incongruent minus neutral. Facilitation effect = neutral minus congruent

*$p < .001$. See text for analysis

$t(39) = 11.188$, $p < .001$; incongruent, $t(39) = 9.085$, $p < .001$; and neutral, $t(39) = 4.020$, $p \leq .001$. More importantly, a significant Stroop interference effect (incongruent minus neutral) was present in the control precue condition, $t(39) = 8.036$, $p < .001$, but was absent in the identity precue condition, $t(39) = 0.127$, $p = .899$. A nonsignificant facilitation effect (neutral minus congruent) was found in the control precue condition, $t(39) = -0.978$, $p = .334$, but was significant in the identity precue condition, $t(39) = -8.334$, $p < .001$.

To quantify support for the null interference effect in the identity condition, we calculated the Bayes factor (BF_{01}) with the null hypothesis as the numerator and the alternative hypothesis as the denominator. The BF_{01} was 5.817, indicating that the null hypothesis is five times more likely than the alternative given the data.[3]

Delta plots

The pattern in the delta plots is supported by separate one-way ANOVAs for interference and facilitation effects in the control and identity precue conditions across the nine quantiles treated as polynomial contrasts. The interference effect in the control precue condition showed a significant linear trend reflecting the positive delta slope, $F(1,39) = 30.160$, $MSe = 368470.820$, $p < .001$, $\eta_p^2 = .436$. In the identity precue condition, the interference effect showed a significant linear trend in the negative direction, reflecting a negative delta, $F(1,39) = 6.312$, $MSe = 69502.589$, $p = .016$, $\eta_p^2 = .139$. For the facilitation effect, the linear trend contrast was flat and non-significant in the control precue condition, $F(1,39) = .050$, $MSe = 504.787$, $p = .824$, $\eta_p^2 = .001$, and in the identity precue condition, $F(1,39) = .452$, $MSe = 5286.946$, $p = .505$, $\eta_p^2 = .011$.

Discussion

The present study demonstrated that distractor priming—precueing the distractor word ahead of the integrated Stroop stimulus—removes the positive delta slope associated with the Stroop interference effect across the whole RT distribution. At the same time, a large Stroop facilitation effect that was constant across the quantiles emerged. In the following section, we discuss the results in terms of "breaking the Gestalt" (La Heij et al., 2001; Lamers & Roelofs, 2007), separating the color and the word in *time*.

Just as the previous studies broke the color-word integration of a Stroop stimulus by removing the color from the stimulus before the offset of the word, we argue that in distractor priming, the onset of the distractor word in monochrome before the onset of the integrated Stroop stimulus disrupts the Gestalt coupling of common fate (i.e., the color and word onsets are not simultaneous), causing participants to no longer see the color and word attributes as belonging to a single integrated object. This, in turn, allows early attentional disengagement from the word and redirects said attention toward the target color to facilitate color naming.

Support for this can be found in the delta plot patterns. We concur with Pratte et al. (2010) and Kinoshita et al. (2017) that the positively sloped delta plot is concordant with the evidence accumulation framework. We argue that this typical positive delta plot is the consequence of the Trojan horse effect (Roelofs & Lamers, 2007): In the integrated Stroop stimulus, spatial or object-based attention results in the irrelevant word being processed alongside the target color. Consequently, the word information inadvertently enters the evidence accumulation process simultaneously alongside the target color information, producing the positive delta plot. Distractor priming breaks the coupling of the word and color attributes in the integrated Stroop stimulus. It allows the early disengagement of attention from the irrelevant word dimension, preventing the distractor word information from simultaneously entering the evidence accumulation process with the target color. As a result, distractor priming removes the positive delta slope associated with the Stroop interference effect.

In our experiment, distractor priming also facilitated color naming for the nonlinguistic neutral stimuli (the row of Xs). We suggest that presenting the object form (XXXX) in monochrome before the colored XXXX breaks the coupling between the object form and color, prompting early attentional disengagement from the object form. This distractor-priming benefit for neutrals was smaller (24 ms) than for incongruent stimuli (137 ms); this is because, for neutral trials, there was no incongruent color-word information that competes with the target color, which, of course, is the cause of the Stroop interference effect.

The large (79 ms) facilitation effect that emerged with distractor priming contrasts with the usually small or nonexistent Stroop facilitation effect, particularly in manual Stroop tasks (MacLeod, 1991). It must be noted that our "breaking the Gestalt" account does not explain this finding: If distractor priming just allows the disengagement of attention from the word form, then there should be no difference between the congruent and incongruent conditions in the distractor-primed condition, contrary to the results found. Other accounts of the distractor-priming benefit cannot explain this finding either. For example, to explain the speedup of color naming of distractor-primed incongruent

[3] Additionally, to investigate whether the elimination of the Stroop interference effect was reliably present at the individual participant level, we adopted the approach of Haaf and Rouder (2019) and plotted the individual effects of Stroop interference in the identity (distractor-primed) conditions. The plots and the summary of the analysis are in Appendix B.

trials, Chao (2011) proposed "active inhibition," in which participants actively inhibit the distractor word when precued. If participants were actively "inhibiting" the distractor word, responses should not be faster for the congruent trials than for the neutral trials in the distractor-primed condition.

The facilitation-dominant pattern and the associated flat delta plot pattern in the distractor-primed condition resemble those found in the primed Stroop task (Kinoshita et al., 2017). One explanation suggested in that paper was that in the primed Stroop task, the word prime is used to predict the response. For example, when presented with the word RED (in monochrome), participants automatically predict the response to be "red," thereby lowering the response threshold for RED. Simulation in that paper showed that a change in response threshold produced a constant effect across the quantiles (i.e., a flat delta plot pattern). Here, in the distractor-primed Stroop condition, the distractor prime may have similarly been used to predict the response by lowering the response threshold, thus producing a large facilitation effect that manifested as a flat delta plot. Additionally, we observed an increase in the size of the facilitation effect over the eight blocks of this experiment (see Appendix C). This pattern is expected from the response threshold account, assuming that participants may have learned to selectively adjust their response thresholds based on the presented precue as they recognized the advantage.

This response-priming process may also have played a role in incongruent trials, given that only a single incongruent color was paired with the distractor (e.g., RED was always displayed in green in the incongruent trials). More specifically, within an evidence accumulation framework that contains four accumulators (one for each color response), when the precue (e.g., RED) is presented, participants may adjust their response thresholds for the congruent color (red) and, if they learn the specific color-word pairings, for the specific incongruent color (green) as well. This would explain the general speedup for congruent and incongruent trials relative to the fixed neutral condition in which no threshold adjustment is possible.

However, this response threshold account based on the learned precue-(incongruent) color pairing alone cannot explain why object-based attention is no longer captured by the distractor word when the integrated Stroop stimulus appears, the main driver of the Stroop interference effect in the standard Stroop.

To conclude, the present study tested the assumption that the integrated nature of the Stroop stimulus, which "traps" the attention to the irrelevant word dimension via the object-based attention, is driving why it is so difficult "not to Stroop." Following Roelofs and Lamers (2007), we predicted that breaking this integration would allow early disengagement of attention from the irrelevant word dimension. We showed that precueing the word can break the integration of the word and color and removes the positively sloped delta plot pattern associated with Stroop interference.

Supplementary information The online version contains supplementary material available at https://doi.org/10.3758/s13423-024-02610-4.

Authors' contributions Not applicable.

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Availability of data and materials The raw data and the output of the statistical analysis from this study can be found on the Open Sciences Framework (https://osf.io/ptm6k/?view_only=d531b2e7a6814ccfbc3b4816a3c35c44).

Code availability Not applicable.

Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

Ethics approval The experiment had the approval of the Macquarie University Human Research Ethics Committee (reference #520231272546753).

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent for publication All individual participants provided additional informed consent to have their deidentified data made available for publication and on the Open Sciences Framework.

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Open practices statement The raw data and the output of the statistical analysis from this study can be found on the Open Sciences Framework (https://osf.io/ptm6k/?view_only=d531b2e7a6814ccfbc3b4816a3c35c44). The study was not preregistered.

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